WEEK 4 REPORT

SPLUNK, WAZUH AND LOG INGESTION

Cybersecurity Internship Weekly Report

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Program	DG Interns Hub
Reporting Period	Week 4
Focus	Splunk , Wazuh & Log Ingestion
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1. INTRODUCTION

Cybersecurity is an important part of protecting computer systems, networks, applications and sensitive information from unauthorized access and cyber threats.

Organizations generate large amounts of security data through operating systems, servers, applications, firewalls and network devices. These systems generate logs that record different activities, including user logins, process execution, network connections and system changes.

However, collecting logs individually is not enough to identify complex security threats. Security teams need a centralized platform to collect, monitor, correlate and analyze security events.

Security Information and Event Management (SIEM) helps organizations perform these activities.

During Week 4 of the DG Interns HUB Cybersecurity Internship, I focused on understanding and implementing SIEM concepts using Splunk Enterprise and Wazuh.

I performed practical activities related to log ingestion, endpoint monitoring, security detection, alert investigation and threat analysis.

This week helped me understand how SOC analysts use SIEM platforms to monitor systems, identify suspicious activities and investigate potential security incidents.

2. OBJECTIVES

The main objectives of this week were:

1. Understand the concept and importance of SIEM.
2. Learn SIEM architecture and log collection methods.
3. Install and configure Splunk Enterprise.
4. Understand Splunk Search Processing Language (SPL).
5. Collect Windows and Linux security logs.
6. Install and configure Wazuh.
7. Monitor endpoint activities using security agents.
8. Understand File Integrity Monitoring (FIM).
9. Perform controlled security testing in a lab environment.
10. Detect brute-force attempts and network scanning activities.
11. Create custom SIEM detection searches and alerts.
12. Investigate and classify security alerts.
13. Understand SOC analyst responsibilities and alert triage.

3. TOOLS AND TECHNOLOGIES USED

The following tools and technologies were used during the practical activities.

Tool / Technology	**Purpose**
Splunk Enterprise	Centralized log searching, monitoring and analysis
Splunk Universal Forwarder	Forwarding endpoint logs to Splunk
Wazuh	Endpoint security monitoring and threat detection
Sysmon	Monitoring Windows process and network activities
Kali Linux	Security testing and network analysis
Ubuntu Linux	Linux server and authentication log collection
Windows	Endpoint security monitoring
Nmap	Network scanning and service discovery
Hydra	Controlled authentication testing
Wireshark	Network traffic analysis
MITRE ATT&CK	Understanding and mapping attacker techniques

4. THEORETICAL LEARNING

During Week 4, I studied eight major theoretical topics.

4.1 Introduction to SIEM

SIEM stands for Security Information and Event Management.

It is a security solution that collects, stores, monitors and analyzes logs from different sources to identify suspicious activities and potential security threats.

Main Functions:

- Centralized log collection.
- Security event monitoring.
- Threat detection.
- Event correlation.
- Alert generation.
- Incident investigation.
- Security reporting.

4.2 Log Ingestion Methods

Log ingestion is the process of collecting log data from different systems and sending it to a centralized monitoring platform.

Important methods include:

- **Agent-Based Collection:** An agent is installed on an endpoint to collect and forward logs.
- **Agentless Collection:** Logs are collected without installing a dedicated endpoint agent, using methods such as Syslog or APIs.
- **API-Based Collection:** Data is collected through application programming interfaces.
- **File Monitoring:** A monitoring tool reads log files and forwards new entries.

4.3 Splunk Search Processing Language (SPL)

SPL stands for Search Processing Language.

It is used in Splunk to search, filter, organize and analyze machine-generated data.

Important SPL commands:

- search
- where
- fields
- table
- stats
- eval
- top

- rare
- dedup
- rex
- regex
- timechart
- eventstats
- iplocation
- lookup

Example:

index=lab EventID=4625 | stats count by User| sort - count

This query searches failed login events and counts them by user.

4.4 Wazuh Architecture

Wazuh is an open-source security monitoring platform used for endpoint monitoring, threat detection and security analysis.

Main Components:

1. Wazuh Agent – Collects security information from endpoints.
2. Wazuh Manager – Processes and analyzes collected events.
3. Wazuh Indexer – Stores and indexes security data.
4. Wazuh Dashboard – Provides a graphical interface for monitoring and investigation.

Other important concepts include decoders, rules and alert severity levels.

4.5 Alert Classification

Security alerts are classified to help analysts determine whether an activity is suspicious.

- True Positive (TP): A real security threat is correctly detected.
- False Positive (FP): A normal activity is incorrectly detected as a threat.
- True Negative (TN): Normal activity is correctly identified as normal.
- False Negative (FN): A real threat is not detected.

Proper alert classification reduces unnecessary investigation and improves security monitoring.

4.6 Brute Force Detection

A brute-force attack is an attempt to gain unauthorized access by repeatedly trying different passwords.

Common indicators include:

- Multiple failed login attempts.
- Repeated authentication requests.
- Attempts from a single suspicious IP address.
- Successful login after multiple failures.

MITRE ATT&CK Technique: T1110 – Brute Force.

4.7 Correlation and Baselines

Event correlation connects related security events from different sources to provide additional investigation context.

A security baseline represents normal system or user behaviour.

Examples:

- Normal login times.
- Common user activities.
- Expected network connections.
- Regular process execution.

Unusual activities can be identified by comparing events against these baselines.

4.8 SOC Tiers and Triage

A Security Operations Center (SOC) monitors and investigates security events.

SOC L1: Performs initial alert monitoring, validation and classification.

SOC L2: Performs deeper investigation and incident analysis.

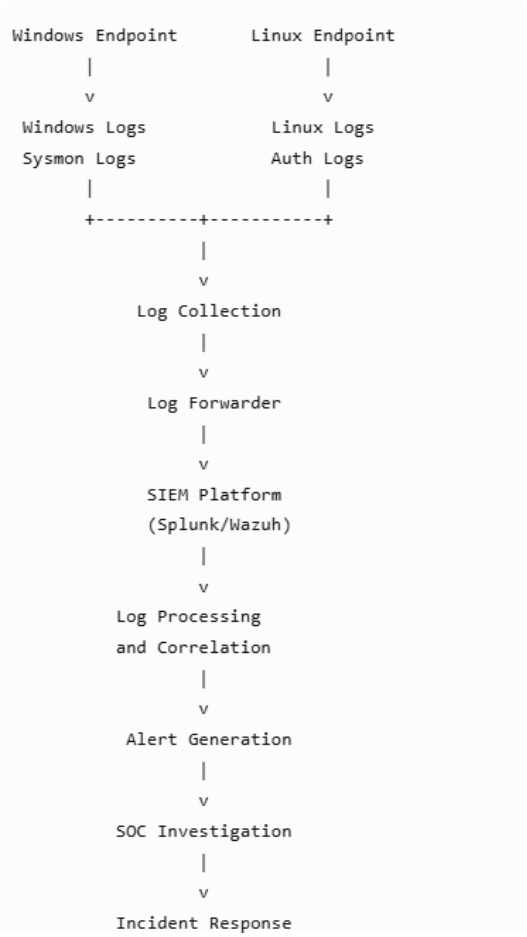
SOC L3: Handles advanced threat hunting, complex incidents and detection improvements.

Alert triage helps security teams prioritize and investigate potential incidents.

5. SIEM ARCHITECTURE AND WORKING

A SIEM platform collects security data from multiple sources and processes it for monitoring and investigation.

Basic SIEM Data Flow



Explanation

1. Endpoints generate system and security logs.
2. Agents or forwarders collect the logs.
3. Logs are transmitted to the SIEM platform.
4. The SIEM processes and analyzes the data.
5. Detection rules identify suspicious activities.
6. Alerts are generated for investigation.
7. SOC analysts investigate and document the results.

6. PRACTICAL IMPLEMENTATION

6.1 TASK 1 – SPLUNK ENTERPRISE INSTALLATION

Objective

To install Splunk Enterprise and prepare a centralized environment for security log collection and analysis.

Activities Performed

1. Downloaded and installed Splunk Enterprise.
2. Configured the initial administrator account.
3. Accessed the Splunk web interface.
4. Explored the Search and Reporting application.
5. Created a custom index named lab.
6. Verified the index configuration.

Understanding Indexes

An index is a storage location where Splunk stores and organizes collected machine data.

Example SPL search: index=lab

This query retrieves events stored in the lab index.

Result

Splunk Enterprise was configured as the centralized log analysis platform for the practical activities.

Evidence to attach:

- Splunk installation screenshot.
- Splunk login or dashboard screenshot.
- Custom index creation screenshot.

6.2 TASK 2 – LOG INGESTION

Objective

To collect Windows and Linux security logs and make them available for searching and investigation in Splunk.

Activities Performed

Windows Log Collection

- Configured Splunk Universal Forwarder.
- Collected Windows Security events.
- Configured Sysmon log collection.
- Forwarded endpoint logs to Splunk.

Linux Log Collection

- Configured Ubuntu authentication log collection.
- Monitored /var/log/auth.log.
- Forwarded authentication events to Splunk.

Important Windows Events

Event ID	Description
4624	Successful login
4625	Failed login
4720	User account created
4732	User added to local security-enabled group
1102	Security audit log cleared
7045	New service installed

Example SPL Search

index=lab

To search for failed Windows logins:

index=lab EventID=4625

Log Ingestion Flow

Windows / Ubuntu

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Security Log Files

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Splunk Forwarder

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Splunk Indexer

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Search and Reporting

Result

Windows Security logs, Sysmon events and Ubuntu authentication logs were collected and made available for SIEM investigation.

Evidence to attach:

- Forwarder configuration.
- Windows log search results.
- Sysmon event search results.
- Ubuntu authentication log search results.

6.3 TASK 3 – WAZUH INSTALLATION AND CONFIGURATION

Objective

To configure Wazuh for endpoint security monitoring and identify suspicious file modifications through File Integrity Monitoring.

Activities Performed

1. Installed Wazuh on Ubuntu.
2. Accessed the Wazuh Dashboard.
3. Connected the Ubuntu endpoint agent.
4. Connected the Windows endpoint agent.
5. Verified agent connectivity.
6. Configured File Integrity Monitoring.
7. Created a test file and modified it.
8. Monitored the generated security alert.

File Integrity Monitoring

File Integrity Monitoring (FIM) is a security feature that monitors files and directories for changes.

It can detect:

- File creation.
- File modification.
- File deletion.
- Changes in file attributes.

Wazuh was used to monitor endpoint activities and identify changes in the configured test files.

Evidence to attach:

- Wazuh Dashboard.
- Connected agent status.
- FIM configuration.
- File modification alert.

6.4 TASK 4 – ATTACK-TO-ALERT PRACTICAL

Objective

To understand how controlled security testing activities generate logs and how SIEM platforms can identify suspicious behaviour.

All testing was performed in my own authorized lab environment.

A. SSH Brute-Force Testing

A brute-force attack attempts to discover valid login credentials through repeated password attempts.

Tool Used: Hydra
Target: Own Ubuntu lab VM
Protocol: SSH

Example controlled lab command:

hydra -l testuser -P passwords.txt ssh://<UBUNTU_LAB_IP> -t 2

Investigation

- Monitored SSH authentication attempts.
- Checked failed authentication events.
- Identified the source IP.
- Reviewed the affected username.
- Searched for related events in SIEM.

Example Ubuntu log location:

/var/log/auth.log

B. Nmap SYN Scan

Nmap is a network scanning tool used to identify open ports and services.

A SYN scan sends TCP SYN packets to identify accessible ports.

Example command against an authorized lab target:

nmap -sS <UBUNTU_LAB_IP>

Investigation

- Observed network scanning activity.
- Identified the scanning source.
- Reviewed the target IP and ports.
- Checked available network logs.
- Investigated related SIEM events.

MITRE ATT&CK Mapping

Activity	Technique
SSH Brute Force	T1110 – Brute Force
Nmap SYN Scan	T1046 – Network Service Discovery

Result

The practical demonstrated how security testing activities can generate observable events that analysts can investigate using SIEM tools.

Evidence to attach:

- Hydra command and output.
- SSH authentication log.
- Nmap scan output.
- Related SIEM search results.

6.5 TASK 5 – SIEM DETECTION ENGINEERING

Objective

To create security detection searches and alerts using SPL and Wazuh.

A. SPL Detection 1 – Failed Login Detection

Purpose: Identify failed login attempts.

```
index=lab EventID=4625 | stats count by User
| sort - count
```

Security Importance: Helps identify repeated authentication failures and possible brute-force attacks.

B. SPL Detection 2 – PowerShell Process Detection

Purpose: Monitor PowerShell process creation using Sysmon.

```
index=lab EventID=1 | search Image="*powershell.exe"
| table _time Computer User Image CommandLine ParentImage
```

Security Importance: Helps analysts investigate suspicious PowerShell execution.

C. SPL Detection 3 – New User Account Detection

Purpose: Identify new account creation.

index=lab EventID=4720

| table _time TargetUserName SubjectUserName Computer

Security Importance: Helps identify unauthorized account creation.

D. SPL Detection 4 – SSH Failed Login Detection

Purpose: Identify repeated SSH authentication failures.

index=lab

("Failed password" OR "Invalid user")

| stats count by src_ip user

| sort - count

Security Importance: Helps identify suspicious SSH login attempts.

E. SPL Detection 5 – Rare Parent-Child Process Detection

Purpose: Identify uncommon parent-child process combinations.

index=lab EventID=1

| stats count by ParentImage Image

| sort count

Security Importance: Helps identify unusual process relationships that may require investigation.

F. Scheduled Alerts

I configured two searches as scheduled alerts.

The purpose of scheduled alerts is to automatically evaluate selected search conditions and notify analysts when the configured conditions are met.

Evidence to attach:

- Alert configuration.
- Alert search results.
- Alert execution evidence.

G. Dashboard Creation

I created a dashboard containing four panels to visualize important security events.

Suggested panel descriptions:

1. Failed login activity.
2. PowerShell process activity.
3. New user account creation.
4. SSH authentication failures.

Evidence to attach: Four-panel Splunk dashboard screenshot.

H. Custom Wazuh Rule

I configured a custom Wazuh detection rule using the local rules configuration.

Configuration file:

/var/ossec/etc/rules/local_rules.xml

Custom rules use rule IDs of 100000 or higher.

I used wazuh-logtest to test the custom rule and verify its detection behaviour.

Result: The practical provided experience in creating and testing custom security detections.

Evidence to attach:

- Custom rule configuration.
- Wazuh logtest output.
- Generated Wazuh alert.

6.6 TASK 6 – SECURITY ALERT INVESTIGATION

Objective

To investigate security alerts, understand suspicious activities and classify findings based on available evidence.

I investigated three security alerts as part of the SOC triage activity.

Alert Investigation Methodology

For every alert, I considered the following questions:

Question	Investigation Details
Who?	Which user or account was involved?
What?	What activity occurred?
When?	When did the event happen?
Where?	Which host or IP address was involved?
Why?	Why was the activity suspicious?

Alert 1 – Authentication Failure

Alert Name: Multiple Failed Login Attempts

- Activity: Repeated authentication failures.
- Log Source: Windows Security or Ubuntu authentication logs.
- Investigation: Checked username, source IP and event timestamps.
- Classification: [Enter TP or FP based on your evidence].
- Recommended Action: Review authentication activity and verify whether the attempts were authorized.

Alert 2 – Suspicious Process Activity

Alert Name: PowerShell Process Creation

- Activity: PowerShell process execution.
- Log Source: Sysmon.
- Investigation: Checked process name, command line, parent process and user.
- Classification: [Enter TP or FP based on your evidence].
- Recommended Action: Review the command and determine whether it was legitimate or suspicious.

Alert 3 – File Integrity Change

Alert Name: File Modification Detected

- Activity: A monitored file was changed.
- Log Source: Wazuh FIM.
- Investigation: Checked the affected file, modification time and related user activity.

- Classification: [Enter TP or FP based on your evidence].
- Recommended Action: Verify whether the modification was authorized and investigate unexpected changes.

Importance of Alert Investigation

Alert investigation helps SOC analysts:

- Identify real security threats.
- Reduce false positives.
- Understand the impact of suspicious activities.
- Collect supporting evidence.
- Recommend appropriate response actions.
- Maintain investigation records.

Note: The final classification and exact investigation findings should match the actual evidence collected during the practical.

TASK 7 – SIEM CHEAT SHEETS

Cheat Sheet 1 – SPL Commands

I prepared a reference table containing 20 commonly used SPL commands.

The table includes:

- SPL command name.
- Meaning of the command.
- Practical example.

These commands help perform searching, filtering, data transformation, event analysis and security investigation.

Cheat Sheet 2 – SIEM Alerts

I prepared a reference table containing 15 common SIEM security alerts.

The table includes:

- Alert name.
- Likely cause.
- First investigation check.

These cheat sheets provide a quick reference for common SOC analyst activities.

7. TRYHACKME LAB ACTIVITIES

I completed two TryHackMe labs to strengthen my theoretical and practical understanding of SIEM and Splunk.

7.1 Lab 1 – Introduction to SIEM

Platform: TryHackMe

Objective: To understand the basic concepts, architecture and importance of SIEM.

Topics Learned:

- Introduction to SIEM.
- Centralized log collection.
- Security monitoring.
- Log analysis.
- Threat detection.
- SIEM use in SOC operations.

Learning Outcome:

I understood how SIEM platforms collect security logs from multiple sources and help analysts identify suspicious activities.



7.2 Lab 2 – Splunk: Exploring SPL

Platform: TryHackMe

Objective: To understand Splunk Search Processing Language and its practical use in security log analysis.

Topics Learned:

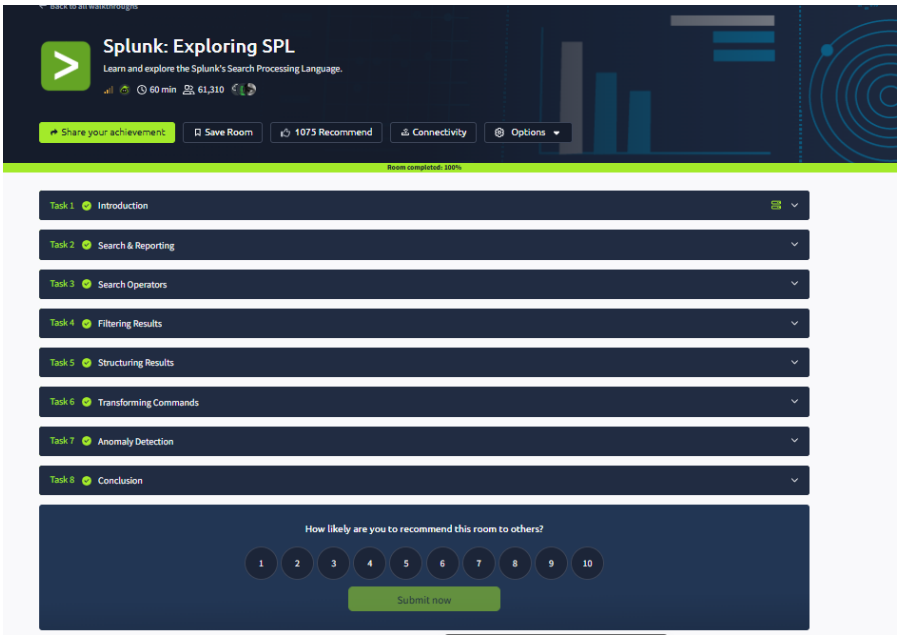
- Splunk Search and Reporting.
- Search operators.
- Filtering commands.
- Structuring commands.
- Transforming commands.
- Event correlation.
- Anomaly detection.

Important SPL Commands Practiced:

search, fields, table, stats, eval, eventstats, top, rare, regex, iplocation and join.

Learning Outcome:

I learned how SPL can be used to search, filter and analyze security events and identify unusual user behaviour.



8. SECURITY DETECTIONS AND MITRE ATT&CK MAPPING

MITRE ATT&CK is a knowledge base that describes common tactics and techniques used by cyber attackers.

During Week 4, I studied how security monitoring activities can be mapped to MITRE ATT&CK techniques.

Security Activity	**MITRE ATT&CK Technique**	**Purpose**
SSH Brute Force	T1110 – Brute Force	Detects repeated credential guessing
Network Scanning	T1046 – Network Service Discovery	Identifies network reconnaissance
PowerShell Execution	T1059.001 – PowerShell	Monitors command and script execution
New Account Creation	T1136 – Create Account	Identifies account creation activity

MITRE mapping provides additional context for understanding attacker behaviour and improving detection coverage.

9. ALERT CLASSIFICATION AND SOC INVESTIGATION

SIEM platforms generate alerts when defined detection conditions are met.

However, not every alert represents a real attack.

SOC analysts must investigate alerts before deciding how to respond.

Types of Alert Classification

- **True Positive:** The alert correctly identifies a real suspicious or malicious activity.
- **False Positive:** The alert identifies normal or authorized activity as suspicious.
- **True Negative:** Normal activity is correctly treated as normal.
- **False Negative:** A real suspicious activity is missed by the detection system.

Basic SOC Investigation Workflow

Security Alert

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Initial Validation

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Log Collection

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Event Analysis

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Alert Classification

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Incident Escalation

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Documentation

Importance of SOC Investigation

- Helps identify real threats.
- Reduces unnecessary alerts.
- Supports incident response.
- Improves detection rules.
- Helps maintain security visibility.

10. CHALLENGES FACED AND SOLUTIONS

During Week 4, I faced different challenges while configuring SIEM platforms and performing practical exercises.

Challenge	**Solution / Approach**
Understanding SIEM architecture	Reviewed the role of agents, managers, indexers and dashboards
Splunk log ingestion configuration	Checked forwarding configuration and log sources
Identifying correct log fields	Examined event details and available fields
Writing SPL queries	Practiced commands and tested search conditions
Connecting Wazuh agents	Checked agent configuration and connectivity
Configuring File Integrity Monitoring	Reviewed monitored directories and file changes
Understanding security alerts	Analyzed timestamps, users, IP addresses and event details
Creating detection rules	Reviewed rule structure and tested using Wazuh logtest
Investigating suspicious activity	Compared related events and reviewed supporting logs

I addressed these challenges through practical testing, reviewing documentation, checking configurations and analyzing generated security events.

11. LEARNING OUTCOMES

After completing Week 4, I gained practical knowledge and understanding of the following areas:

1. SIEM concepts and architecture.
2. Centralized security log collection.
3. Splunk Enterprise installation and configuration.
4. Splunk Search Processing Language.
5. Windows Security and Sysmon log analysis.
6. Linux authentication log monitoring.
7. Wazuh endpoint monitoring.
8. File Integrity Monitoring.
9. Brute-force detection.
10. Network scanning detection.
11. Custom security detection rules.
12. Scheduled alerts and dashboards.
13. MITRE ATT&CK technique mapping.
14. SOC alert investigation and classification.
15. Basic incident response and documentation.

12. CONCLUSION

Week 4 of the DG Interns HUB Cybersecurity Internship provided valuable practical knowledge about Security Information and Event Management.

Throughout the week, I learned how SIEM platforms collect, process and analyze security logs from Windows and Linux endpoints.

I gained hands-on experience with Splunk Enterprise and Wazuh, including log ingestion, endpoint monitoring, File Integrity Monitoring, SPL searches, scheduled alerts and custom detection rules.

I also performed controlled security testing using Hydra and Nmap to understand how suspicious activities generate observable security events.

The alert investigation exercises helped me understand how SOC analysts validate alerts, identify suspicious behaviour, classify findings and document investigation results.

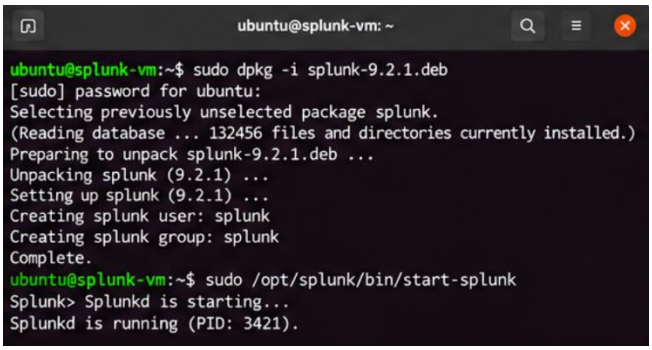
Overall, this week improved my practical understanding of security monitoring, threat detection, log analysis and SOC operations.

The knowledge and skills gained during this internship will help me continue developing my cybersecurity and SOC analyst capabilities.

13. SCREENSHOTS AND PRACTICAL EVIDENCE

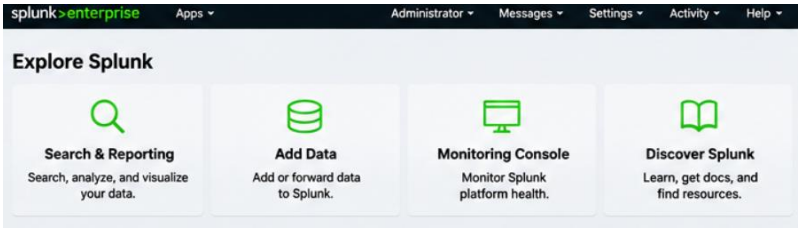
Attach actual screenshots under the following categories.

13.1 Splunk Enterprise



Splunk installation.





Splunk login and dashboard.



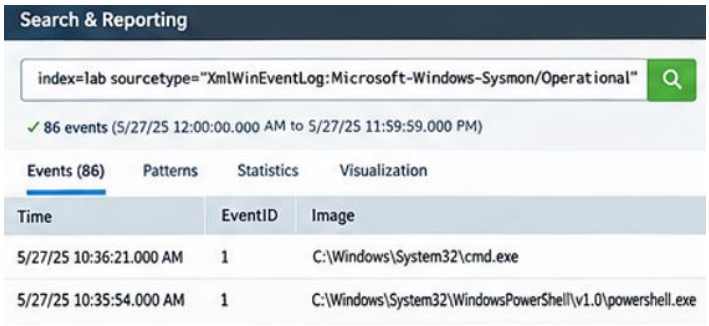
Custom index creation.

13.2 Log Ingestion

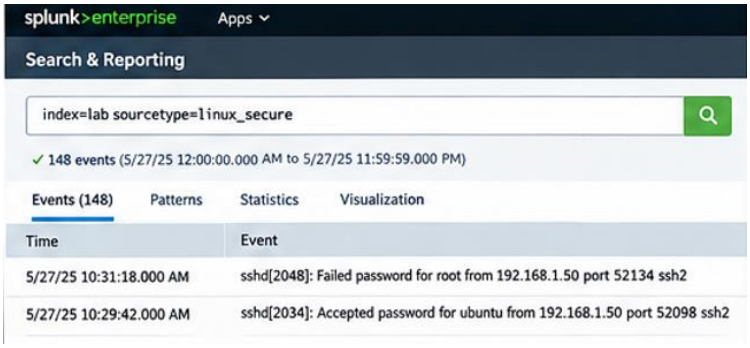
1. Install splunk



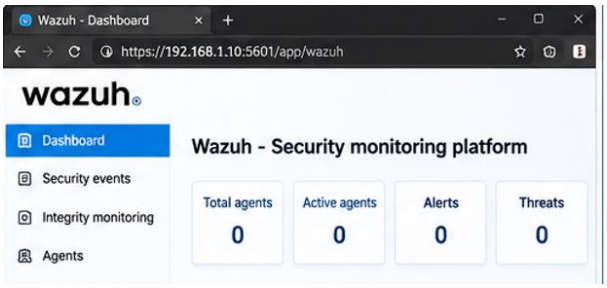
2. Sysmon event



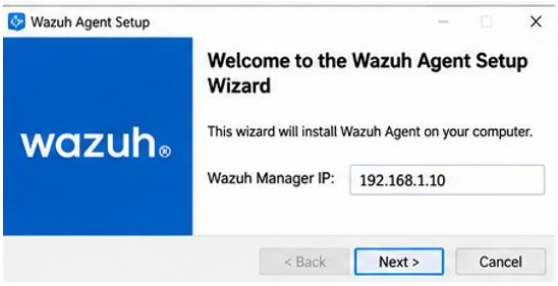
3. Successful log search results



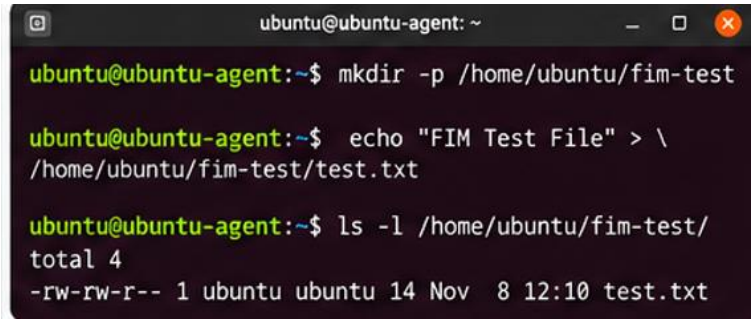
13.3 Wazuh



Wazuh Dashboard.



Windows agent status.



File Integrity Monitoring configuration.



File modification alert.

13.4 Attack-to-Alert Practical

```
kali@kali:~$ hydra -l ubuntu -P /usr/share/wordlists/rockyou.txt \
ssh://192.168.1.20 -t 4
Hydra v9.5 (c) 2024 by van Hauser/THC
[DATA] attacking ssh://192.168.1.20:22/
[22][ssh] host: 192.168.1.20   login: ubuntu   password: ubuntu123
1 of 1 target successfully completed, 1 valid password found
kali@kali:~$
```

Hydra testing command.

```
ubuntu@ubuntu:~$ sudo tail -n 5 /var/log/auth.log
Nov  8 12:38:21 ubuntu sshd[2456]: Failed password
for ubuntu from 192.168.1.50 port 49716 ssh2
Nov  8 12:38:23 ubuntu sshd[2456]: Failed password
for ubuntu from 192.168.1.50 port 49718 ssh2
Nov  8 12:38:25 ubuntu sshd[2456]: Failed password
for ubuntu from 192.168.1.50 port 49720 ssh2
```

SSH failed login logs.

```
kali@kali:~$ nmap -sS 192.168.1.20
Starting Nmap 7.94 ( https://nmap.org ) at 2026-11-08 12:45 IST
Nmap scan report for 192.168.1.20
Host is up (0.00046s latency).
PORT       STATE     SERVICE
22/tcp     open      ssh
80/tcp     open      http
Nmap done: 1 IP address (1 host up)
```

Nmap SYN scan command.

Alerts

Alert Name	Severity	Triggered Time
SSH Brute Force Attempt	High	Nov 8, 12:38 PM
Port Scan Detected (Nmap)	Medium	Nov 8, 12:45 PM

SIEM search results.

13.5 Detection Engineering

splunk>enterprise Apps

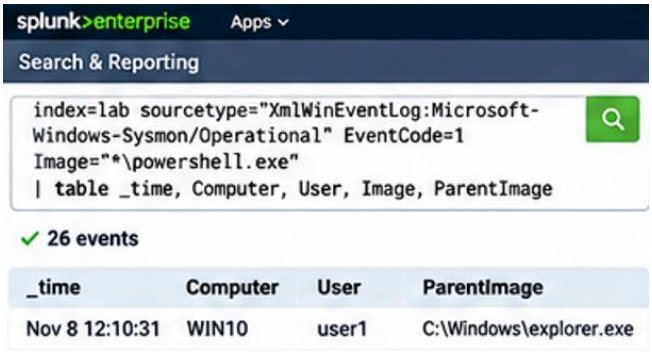
Search & Reporting

```
index=lab sourcetype=WinEventLog:Security
EventCode=4625
| stats count by Account_Name, src_ip
```

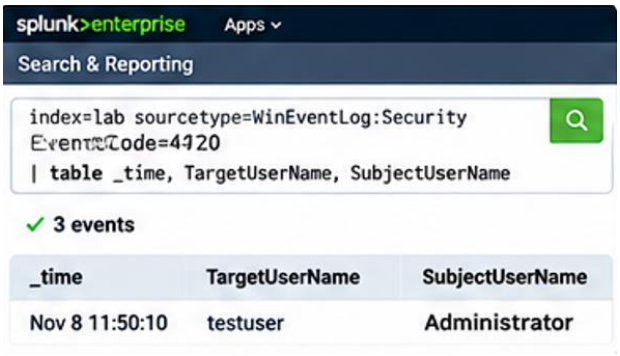
✓ 84 events (last 24 hours)

Account_Name	src_ip	count
admin	192.168.1.50	42
test	192.168.1.60	17

Failed login detection.



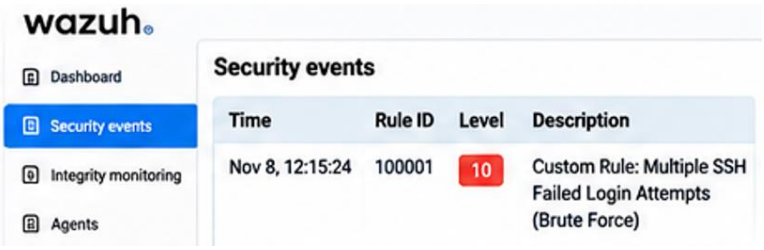
PowerShell process detection.



New user detection.

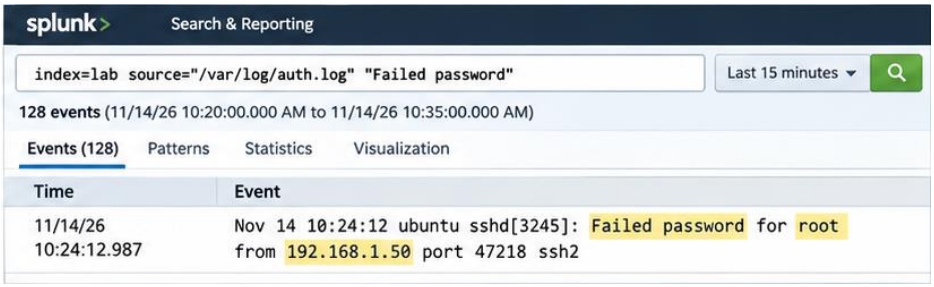


SSH failed login detection.

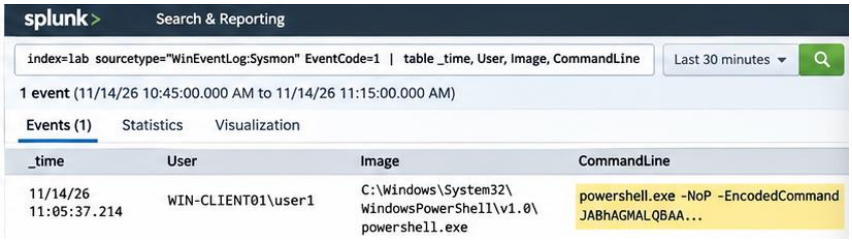


Wazuh logtest output.

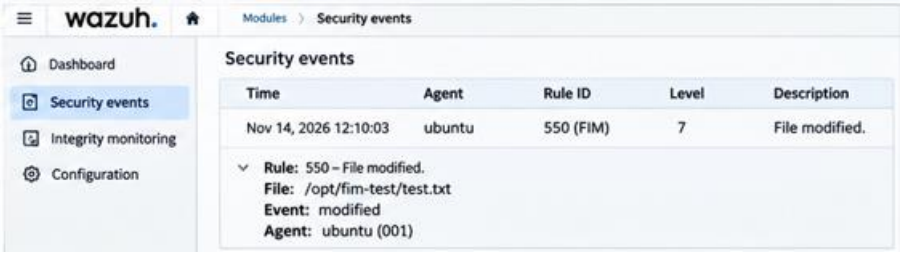
13.6 Alert Investigation



Alert investigation 1



Alert investigation 2.



Alert investigation 3.

13.7 Cheat Sheets

- SPL Commands Cheat Sheet.

No.	SPL Command	Meaning	Practical Example
1	`search`	Searches specific events	`index=lab | search EventID=4625`
2	`where`	Filters results using conditions	`index=lab | stats count by User | where count>5`
3	`fields`	Selects required fields	`index=lab | fields User EventID`
4	`table`	Displays data in table format	`index=lab | table _time User EventID`
5	`stats`	Calculates statistics	`index=lab | stats count by User`
6	`top`	Displays most common values	`index=lab | top User`
7	`rare`	Displays least common values	`index=lab | rare User`
8	`eval`	Creates or modifies fields	`index=lab | eval Status="Failed"`
9	`dedup`	Removes duplicate field values from results	`index=lab | dedup User`
10	`sort`	Sorts search results	`index=lab | sort - _time`
11	`head`	Displays first results	`index=lab | head 10`
12	`tail`	Displays last results	`index=lab | tail 10`
13	`reverse`	Reverses result order	`index=lab | reverse`
14	`rex`	Extracts fields using regex	`index=lab | rex "user=(?<username>\w+)"`
15	`regex`	Filters using regular expressions	`index=lab | regex User="admin.*"`
16	`eventstats`	Adds statistics to original events	`index=lab | eventstats count by User`
17	`timechart`	Creates time-based statistics	`index=lab | timechart count`
18	`iplocation`	Adds geographical IP information	`index=lab | iplocation SourceIp`
19	`lookup`	Enriches events using lookup tables	`index=lab | lookup users.csv User OUTPUT Department`
20	`join`	Combines related search results	`index=lab | join User [search index=lab EventID=4625]`

- **SIEM Alerts Cheat Sheet.**

No.	SIEM Alert	Likely Cause	First Investigation Check
1	Multiple Failed Logins	Brute-force attack or wrong password	Check username, source IP and failure count
2	Successful Login After Failures	Possible compromised account	Check previous failures and successful login source
3	SSH Brute Force	Repeated SSH login attempts	Check `/var/log/auth.log` and source IP
4	New User Account Created	Unauthorized or legitimate account creation	Check Event ID 4720 and account name
5	Suspicious PowerShell Execution	Malicious script or administrative activity	Check command line and parent process
6	Nmap Port Scanning	Network reconnaissance or authorized scan	Check source IP and scanned ports
7	File Integrity Change	Unauthorized or expected file modification	Check filename, modification time and user
8	Suspicious Process Creation	Malware or unusual application execution	Check process name, path and parent process
9	Privilege Escalation	Unauthorized privilege changes	Check account privileges and related events
10	Security Log Cleared	Evidence removal or authorized maintenance	Check Event ID 1102 and responsible account
11	Antivirus Detection	Malware or potentially unwanted software	Check detected file, threat name and action
12	Unusual Login Location	Compromised credentials or VPN usage	Check source IP, country and login time
13	Password Spraying	Common password attempts across multiple accounts	Check failed attempts across usernames
14	New Service Installed	Legitimate software installation or persistence	Check Event ID 7045 and service path
15	Unusual Outbound Connection	Suspicious communication or normal application traffic	Check destination IP, port and process

14. REFERENCES

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2. Wazuh Documentation – https://documentation.wazuh.com/
3. MITRE ATT&CK Framework – https://attack.mitre.org/
4. TryHackMe – https://tryhackme.com/
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